

Strengthening the DORAEMON “Tap-and-Charge” System Proposal

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1 Introduction and Concept Overview

DORAEMON (Detached On-demand Rapid Absorption Energy Mechanism for Optimized Networks) is a proposed wireless “tap-and-charge” system that aims to transfer energy almost instantaneously to a device via brief contact (on the order of 50–200 ms). The key innovation is decoupling the energy transfer phase from the energy usage phase: a high-power wireless burst rapidly **fills an on-board energy store** (a supercapacitor “energy cavity”), which then discharges over time to power or recharge the device’s battery. This concept addresses the mobility constraints of conventional wireless charging (which typically requires continuous coupling) by enabling quick energy injections during momentary contact [?].

To evaluate and solidify this idea, we need a rigorous theoretical framework covering:

- **Electromagnetic coupling** efficiency during the short power transfer burst (how to achieve high power transfer in a brief contact).
- **Rapid energy absorption** dynamics in the intermediate storage (supercapacitor) including limits from circuit physics (capacitance, ESR, leakage).
- **Optimal discharge control** into the battery to maximize charge speed while preserving battery health.

We examine each aspect with detailed math and physics to ensure feasibility. We also outline how **software simulations** can validate each part of the system, allowing experiments “in code” before any hardware testing.

2 Electromagnetic Coupling Theory for Instant Energy Transfer

2.1 Mutual Inductance and Resonant Coupling

At the core of the wireless link are two resonant coils—one in the transmitter and one in the device—that form a coupled inductive system. The **mutual inductance** M between the coils quantifies how effectively magnetic flux from the transmit coil links to the receive coil. For given coil geometries, M can be calculated by integrating the magnetic field over the coil areas (Biot–Savart law). A standard form is

$$M = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\ell_1 \cdot d\ell_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|}, \quad (1)$$

where the line integrals run over the wire paths of coils C_1 and C_2 . The **coupling coefficient** is $k = M/\sqrt{L_1 L_2}$. In a representative prototype, coils are tuned to $f_0 = 6.78$ MHz (ISM band) achieving $k \approx 0.42$ at ~ 12 mm separation. Strong coupling is crucial to transfer large power in a short time.

Resonant operation: Each coil is paired with a capacitor (C_1, C_2) to form resonant circuits. At $\omega_0 = 2\pi f_0$, the coils exchange energy efficiently via the oscillating magnetic field; in the gap, a time-harmonic field can be written as $\mathbf{H}(\mathbf{r}, t) = \Re\{\mathbf{H}_0(\mathbf{r})e^{j\omega_0 t}\}$. With both coils tuned to ω_0 even as load changes, energy transfer is maximized. Detuning $\Delta\omega$ reduces efficiency approximately by a Lorentzian factor $1/(1 + (\Delta\omega \tau)^2)$ where τ depends on the coil Q factors.

2.2 Power Transfer Efficiency and Optimal Load

Using a two-port RLC model with parasitic resistances R_1 and R_2 and load R_L , efficiency depends on $k^2 Q_1 Q_2$ with $Q_i = \omega_0 L_i / R_i$ [?]. Under optimal matching, a widely used closed form for the **maximum efficiency** is

$$\eta_{\max} = \frac{k^2 Q_1 Q_2}{(1 + \sqrt{1 + k^2 Q_1 Q_2})^2}. \quad (2)$$

For example, with $k^2 Q_1 Q_2 \approx 0.42^2 \times 200 \times 150 \approx 5290$, one finds $\eta_{\max} \approx 94.7\%$, rivaling wired charging. The optimal effective load is on the order of

$$R_{L,\text{opt}} \approx R_2 \sqrt{1 + k^2 Q_1 Q_2}, \quad (3)$$

balancing dissipation and transfer. Operating near the efficiency optimum is preferable to extracting peak power (which often incurs higher coil losses).

Maximum power vs. maximum efficiency. The R_L that maximizes delivered power is generally not the one that maximizes efficiency. In the heavy-load limit (small R_L), coil currents (and copper losses) increase disproportionately, while in the light-load limit (large R_L) efficiency tends to 100% but delivered power vanishes. With limited contact time, harvesting more *net* energy favors operating near the efficiency optimum rather than brute-forcing instantaneous power.

2.3 Meeting the 50 ms Transfer Window

At 6.78 MHz, 50 ms spans roughly 3.4×10^5 cycles—ample for steady-state transfer. The rise time of energy exchange is governed by the damping and coupling; a commonly used estimate is

$$\tau \approx \frac{2Q_1 Q_2}{Q_1 + Q_2} \frac{1}{\omega_0}, \quad (4)$$

which is $\mathcal{O}(\mu\text{s})$ for $Q \sim 150$ – 200 at 6.78 MHz. Hence nearly the entire 50 ms contact can be used efficiently. Power-source and thermal constraints still bound the instantaneous pulse power, but the short duty greatly eases thermal stress (e.g., $100 \text{ W} \times 50 \text{ ms} = 5 \text{ J}$).

Alignment and stability. Because contact time is short, mechanical or magnetic alignment aids should ensure k remains high ($\gtrsim 0.4$). Coil shapes and ferrites can provide tolerance to small misalignments.

3 Rapid Energy Absorption in the Supercapacitor Bank

3.1 Supercapacitor Charging Dynamics and Energy Equation

During contact, the receiver must **capture and store** energy quickly in a supercapacitor bank. Let $C_{\text{eff}}(V_c)$ denote the effective (possibly voltage-dependent) capacitance, R_{ESR} the equivalent series resistance, and $I_{\text{leak}}(V_c, T)$ the leakage current. The dynamics and energy are

$$\frac{dV_c}{dt} = \frac{I_{\text{in}}(t) - I_{\text{leak}}(V_c)}{C_{\text{eff}}(V_c)}, \quad (5)$$

$$E_{\text{stored}}(t) = \int_0^{V_c(t)} C_{\text{eff}}(V) V dV. \quad (6)$$

If $C_{\text{eff}}(V)$ is modeled as $C_0 [1 - \alpha(V/V_{\text{rated}}) + \beta(V/V_{\text{rated}})^2]$, then

$$E_{\text{stored}}(V_c) = C_0 \left[\frac{1}{2} V_c^2 - \frac{\alpha}{3} V_{\text{rated}} V_c^3 + \frac{\beta}{4} V_{\text{rated}}^2 V_c^4 \right]. \quad (7)$$

Instantaneous power balance after rectification is

$$P_{\text{in}} = V_c I_{\text{cap}} + I_{\text{in}}^2 R_{\text{ESR}} + V_c I_{\text{leak}}, \quad I_{\text{cap}} = C_{\text{eff}} \dot{V}_c. \quad (8)$$

For a 50 ms pulse, leakage is negligible; the dominant loss is $I^2 R_{\text{ESR}}$ when V_c is small. The **instantaneous storage efficiency** is

$$\eta_{\text{storage}}(I) = \frac{V_c I}{V_c I + I^2 R_{\text{ESR}}} = \frac{1}{1 + \frac{I R_{\text{ESR}}}{V_c}}. \quad (9)$$

Illustrative example. For $C = 50$ F, $R_{\text{ESR}} = 10$ m Ω , charging at 100 A for 50 ms yields $\Delta V \approx I \Delta t / C = 0.1$ V and $E_{\text{stored}} \approx \frac{1}{2} C V^2 = 0.25$ J, while ESR loss is $I^2 R \Delta t = 5$ J; efficiency $< 5\%$. This motivates **not** brute-forcing current when V_c is low.

Governing relations (summary). For clarity during implementation:

- **Charge balance:** $I_{\text{in}} = I_{\text{cap}} + I_{\text{leak}}$, with $I_{\text{cap}} = C_{\text{eff}}(V_c) \dot{V}_c$.
- **Energy:** $E_{\text{stored}}(t) = \int_0^{V_c(t)} C_{\text{eff}}(V) V dV$ (reduces to $\frac{1}{2} C V_c^2$ if C is constant).
- **Power balance:** $P_{\text{in}} = V_c I_{\text{cap}} + I_{\text{in}}^2 R_{\text{ESR}} + V_c I_{\text{leak}}$.

3.2 Optimal Charging Profile for Maximum Energy Capture

A short-horizon optimal control problem maximizes $E_{\text{stored}}(T_{\text{charge}})$ over $I(t)$ subject to $0 \leq I(t) \leq I_{\text{max}}$ and safety constraints. Quasi-static reasoning suggests moderating I at low V_c and increasing it as V_c rises. A leakage-aware stationary condition sometimes quoted is

$$I \approx \frac{V_c I_{\text{leak}}(V_c)}{R_{\text{ESR}}}, \quad (10)$$

but for 50 ms windows where leakage is negligible this pushes $I \rightarrow 0$, conflicting with time limitations. Practical near-optimal heuristics are:

- **Initial phase (low V_c):** use moderate I to avoid I^2R loss dominance.
- **Middle phase:** ramp toward $I_{\max}$ as V_c increases and efficiency improves.
- **End of contact:** taper if nearing a cap voltage limit or disconnection.

These profiles can be found numerically (e.g., with `fmincon` or dynamic programming) and consistently yield high capture ($\gtrsim 90\%$) given sufficiently low ESR (paralleled cells to reduce R_{ESR}) and adequate matching.

Thermal aspect. Single 50 ms pulses deposit only a few joules of heat, leading to modest temperature rise; repeated taps may need cooldown or heat-sinking.

3.3 Example of Feasibility—Real-World Analogues

ABB’s *flash-charging bus* delivers ~ 400 kW over ~ 15 s at stops[?]. Some Chinese buses use supercapacitor banks for ~ 30 s top-ups[?]. On smaller scales, wearables leverage supercaps for rapid top-ups (e.g., a 1 F cap charged near 1 s at a few watts)[?]. Our goal of 5–20 J in 50 ms (100–400 W) is aggressive but within comparable regimes if the system is tightly tuned.

4 Optimal Discharge and Battery Charging Theory

After detachment, the supercapacitor discharges into the battery over minutes (e.g., 30+) in a **two-stage** process: Stage 1 (50 ms) cap charge; Stage 2 (minutes) cap-to-battery discharge, allowing the battery to see normal fast-charge conditions.

4.1 Battery Model for Charging

We adopt a Randles-type model: an OCV source $\text{OCV}(\text{SOC})$ in series with R_{int} and an RC branch:

$$V_{\text{bat}}(t) = \text{OCV}(\text{SOC}(t)) + I_{\text{bat}}(t) R_{\text{int}} + V_{RC}(t), \quad (11)$$

$$\tau_{RC} \dot{V}_{RC}(t) + V_{RC}(t) = I_{\text{bat}}(t) R_{RC}. \quad (12)$$

Constraints include temperature (via a lumped thermal model), maximum C-rate, and terminal voltage.

4.2 Optimal Current Profile for Battery Charging

Industrial fast charging typically follows CC–CV: **max current until voltage limit**, then hold voltage while current tapers. Pontryagin-style optimal control studies corroborate a “bang–constraint” policy under voltage/thermal constraints[?]. A practical implementation uses a DC–DC that sources up to $I_{\max}$ initially, then tapers as battery voltage approaches the limit or as the cap depletes. The passive cap–battery dynamics can mimic CC-to-CV transitions naturally.

5 Simulation Framework and Verification via Code

We recommend validating each block in code before hardware:

- **Electromagnetics:** FEM (e.g., COMSOL/Maxwell) or circuit models with mutual inductance; sweep k vs alignment, spacing, and detuning.

- **Supercapacitor:** integrate $\dot{V}_c$ with ESR loss; compare constant-current vs tapered profiles; optimize $I(t)$ numerically.
- **Battery discharge:** simulate DC–DC controlled CC–CV into an OCV–RC battery; include a simple thermal rise model.
- **Integrated:** co-simulate RF source + mutual inductor + rectifier + DC–DC + cap + battery; probe edge cases (shorter taps, higher ambient, detuning).

These “in silico” prototypes enable rapid iteration on coil dimensions, capacitor sizing, converter ratings, and control laws.

6 Feasibility and Recommendations

- **High-performance components:** litz-wire coils, careful shielding/ferrites, supercaps with milliohm-level ESR (parallel to reduce ESR).
- **Safety/compliance:** 6.78 MHz ISM operation with strong near-field coupling; add interlocks and object detection.
- **Alignment:** mechanical/magnetic aids to keep k high during the tap.
- **Scaling expectations:** taps are immediately useful for low/medium power devices; EV-scale requires multiple taps or larger pulses.
- **Architectures:** parallel-charge/series-discharge supercap reconfiguration to raise discharge voltage where needed.
- **Adaptive control:** learn optimal current waveforms; adaptive matching to keep resonance optimal under variations.
- **Multi-objective modes:** “boost” vs “eco” to trade efficiency and energy when taps are shorter.

Final verdict. The DORAEMON tap-and-charge concept is **theoretically sound and feasible** with careful engineering. Resonant coupling supports high-efficiency short bursts; supercaps can absorb energy efficiently with shaped pulses; batteries can be charged safely off-line. Proceed with full simulations and a small prototype to validate capture efficiency and end-to-end energy delivery.

Recommendation. Implement the end-to-end simulation stack, then prototype a small receiver (e.g., 50 F at 12–16 V) that demonstrates a 50 ms capture followed by a 30 min CC–CV discharge into a single-cell Li-ion, verifying thermal and efficiency targets.

Sources

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